

Advanced (Habanero)

Q1. Given $f(x) = x^2$ and $g(x) = 2x - 1$, find $f(g(x))$.

- (A) $4x^2 - 4x + 1$
- (B) $4x - 1$
- (C) $2x^2 - x$
- (D) $4x^2 - 4$
- (E) $2x^2 - 2x + 1$

Q2. Given $f(x) = \frac{1}{x}$ and $g(x) = x - 2$, find the domain of $f(g(x))$.

- (A) $(-\infty, 2) \cup (2, \infty)$
- (B) $(-\infty, 2)$
- (C) $(-\infty, 0) \cup (0, 2) \cup (2, \infty)$
- (D) $[2, \infty)$
- (E) $(-\infty, 0) \cup (2, \infty)$

Q3. If $f(x) = x + 2$ and $g(x) = 2x - 3$, what is $g(f(x))$?

- (A) $2x - 1$
- (B) $2x + 1$
- (C) $4x - 3$
- (D) $2x - 3$
- (E) $4x + 1$

Q4. Given $f(x) = x^2$ and $g(x) = \sqrt{x}$, find $f(g(x))$.

- (A) x^4
- (B) x^2
- (C) $\sqrt{x}$
- (D) x
- (E) x^3

Q5. Given $f(x) = x^2 - 2x + 1$ and $g(x) = x - 3$, find the value of $f(g(0))$.

- (A) 1
- (B) 4
- (C) 0
- (D) 2
- (E) 16

Q6. Given $f(x) = x^3$ and $g(x) = \frac{1}{x}$, find the domain of $g(f(x))$.

- (A) $(-\infty, 0) \cup (0, \infty)$
- (B) $(-\infty, 0)$
- (C) $(-\infty, 0) \cup (0, 1)$
- (D) $[0, \infty)$
- (E) $(-\infty, -1) \cup (1, \infty)$

Q7. If $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, find $f(g(x))$.

- (A) x
- (B) $x^2 - 2$
- (C) x^2
- (D) $x + 1$
- (E) $x - 1$

Q8. Given $f(x) = |x|$ and $g(x) = -x$, find $f(g(x))$.

- (A) x
- (B) $-x$
- (C) $|x|$
- (D) x^2
- (E) $-|x|$

Open Ended Questions (FRQ)

Q9. Find the value of $f(g(x))$ given $f(x) = 2x + 1$ and $g(x) = x - 4$. Show all steps and simplify your answer.

Q10. If $f(x) = x^2 - 3x + 2$ and $g(x) = 2x - 1$, determine the domain of $f(g(x))$. Explain your reasoning and provide the final domain.

Chef's Tips

- Tip 1: Emphasize the importance of understanding function composition
- Tip 2: Review domain restrictions for composite functions
- Tip 3: Provide real-world examples to illustrate the application of composite functions